

Configure IP-Based Routing Policy in Amazon Route 53

Summary:-

In this lab, an IP-Based Routing Policy is configured in Amazon Route 53. CIDR Collections are created to associate specific source IP address ranges with different DNS responses. DNS records are then configured to return different IP addresses based on the source IP address of the DNS resolver. The routing policy is verified using both **nslookup** and the **Test Record** feature.

Let's Start:-

- Create a CIDR Collection in Amazon Route 53.
- Add CIDR locations for different source IP ranges.
- Configure IP-Based Routing Policy records.
- Associate each record with a CIDR location.
- Verify the routing policy using DNS queries.

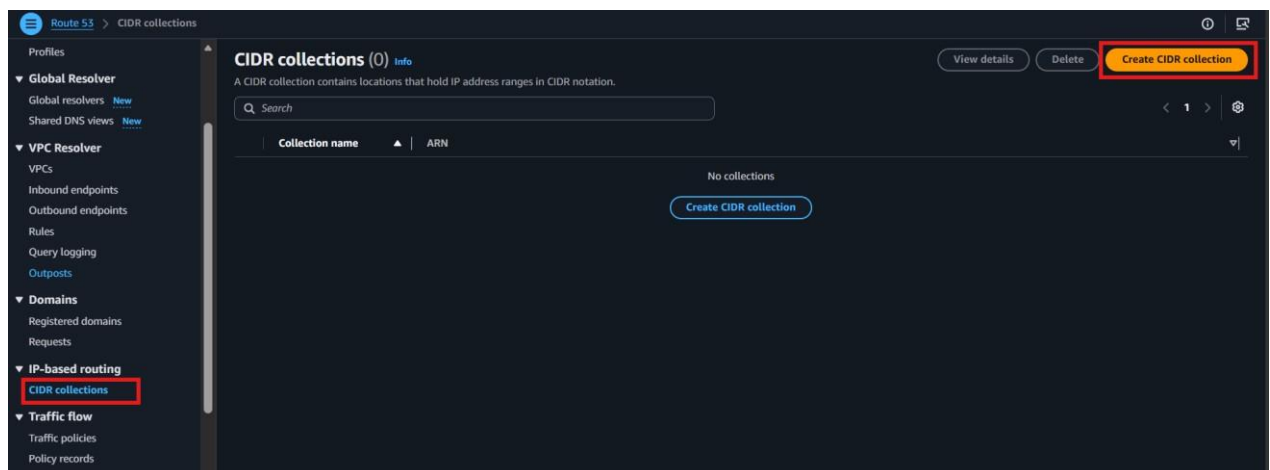
Lab Steps:-

Step 1 – Verify the Environment

- Open the **Amazon Route 53** console.
- Select an existing **Public Hosted Zone**.
- This step prepares the required Route 53 resources for the IP-Based Routing Policy.

Step 2 – Create a CIDR Collection

- Open the **Amazon Route 53** console.
- Select **CIDR Collections**.
- Click **Create CIDR Collection**.



- Enter the following:
 - **Collection Name: C1**

Route 53 > CIDR collections > Create CIDR collection

Create CIDR collection Info

To create a CIDR collection, you must create at least one CIDR location with CIDR blocks. All record sets with the same record set name and type must reference the same collection.

Details
Collection name
C1
Valid characters: a-z, 0-9, -, _

Create CIDR locations
To create a CIDR collection, you must create at least one CIDR location with CIDR blocks.
[Add another location](#)
▼ CIDR location 1
[Remove location](#)
CIDR location
Gujrat
Valid characters: a-z, 0-9, -, _
CIDR blocks
List of IP addresses from /0 to /24 for IPv4 and /0 to /48 for IPv6.
119.0.0.0/8

- This step creates a collection to store CIDR locations.

Step 3 – Add CIDR Locations

- Add the following CIDR locations:
 - **Location: Gujarat**
 - **CIDR: 119.0.0.0/8**

Route 53 > CIDR collections > Create CIDR collection

Create CIDR collection Info

To create a CIDR collection, you must create at least one CIDR location with CIDR blocks.

Details
Collection name
C1
Valid characters: a-z, 0-9, -, _

Create CIDR locations
To create a CIDR collection, you must create at least one CIDR location with CIDR blocks.
[Add another location](#)
▼ CIDR location 1
[Remove location](#)
CIDR location
Gujrat
Valid characters: a-z, 0-9, -, _
CIDR blocks
List of IP addresses from /0 to /24 for IPv4 and /0 to /48 for IPv6.
119.0.0.0/8
▼ CIDR location 2
CIDR location
Maharashtra
Valid characters: a-z, 0-9, -, _
CIDR blocks
200.0.0.0/8
[Add another location](#)

- **Location: Maharashtra**
- **CIDR: 200.0.0.0/8**

Route 53 > CIDR collections > Create CIDR collection

CIDR location 2 Remove location

CIDR location
 Maharashtra
 Valid characters: a-z, 0-9, -

CIDR blocks
 List of IP addresses from /0 to /24 for IPv4 and /0 to /48 for IPv6.
 200.0.0.0/8

Add another location

Cancel Create CIDR collection

- Click **Create CIDR Collection**.
- This step associates source IP ranges with different CIDR locations.

Step 4 – Create the Gujarat Record

- Open the existing **Public Hosted Zone**.

Route 53 > Hosted zones

Hosted zones (1) View details Edit Delete Create hosted zone

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Hosted zone name	Type	Created by	Record count	Description	Hosted zone ID
cloudservicesdemo.in	Public	Route 53	2	-	Z02348701U0CZJKX6V4XE

- Click **Create Record**.

Route 53 > Hosted zones > cloudservicesdemo.in

Public cloudservicesdemo.in Delete zone Test record Configure query logging

Hosted zone details Edit hosted zone

Records (2) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (2) info Delete record Import zone file Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Record name	Type	Routing policy	Alias	Value/Route traffic to	TTL (s)	Health status	Evaluation status	Record ID
cloudservice...	NS	Simple	No	ns-1598.awsdns-07.co.uk. ns-16.awsdns-02.com. ns-925.awsdns-51.net. ns-1212.awsdns-23.org.	1,72,800	-	-	-
cloudservice...	SOA	Simple	No	ns-1598.awsdns-07.co.uk. a...	900	-	-	-

- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** 1.1.1.1
 - **Routing Policy:** IP-Based
 - **CIDR Collection:** C1

- **Location:** Gujarat
- **Record ID:** R1

The screenshot shows the AWS Route 53 'Create record' interface. The breadcrumb trail is 'Route 53 > Hosted zones > cloudservicesdemo.in > Create record'. The form for 'Record 1' includes the following fields:

- Record name:** learn (with a red box around the text)
- Record type:** A - Routes traffic to an IPv4 address and some AWS resources (with a red box around the dropdown text)
- Value:** 1.1.1.1 (with a red box around the text)
- Routing policy:** IP-based (with a red box around the dropdown text)
- Record ID:** R1 (with a red box around the text)
- Location:** Gujarat (selected in the dropdown, with a red box around the text)

- Click **Create Records**.
- This step returns the Gujarat endpoint for the configured source IP range.

Step 5 – Create the Maharashtra Record

- Click **Create Record**.
- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** 2.2.2.2
 - **Routing Policy:** IP-Based
 - **CIDR Collection:** C1
 - **Location:** Maharashtra
 - **Record ID:** R2

The screenshot shows the AWS Route 53 'Create record' interface for a second record. The breadcrumb trail is 'Route 53 > Hosted zones > cloudservicesdemo.in > Create record'. The form for 'Record 2' includes the following fields:

- Record name:** learn (with a red box around the text)
- Record type:** A - Routes traffic to an IPv4 address and some AWS resources (with a red box around the dropdown text)
- Value:** 2.2.2.2 (with a red box around the text)
- Routing policy:** IP-based (with a red box around the dropdown text)
- Record ID:** R2 (with a red box around the text)
- Location:** Maharashtra (selected in the dropdown, with a red box around the text)

The 'Add another record' button at the bottom right is highlighted with a red box.

- Click **Create Records**.

- This step returns the Maharashtra endpoint for the configured source IP range.

Step 6 – Create the Default Record

- Click **Create Record**.
- Configure the following:
 - **Record Name:** learn
 - **Record Type:** A
 - **Value:** 3.3.3.3
 - **Routing Policy:** IP-Based
 - **Location:** Default
 - **Record ID:** R3

The screenshot shows the AWS Route 53 'Create record' page. The breadcrumb trail at the top indicates the path: Route 53 > Hosted zones > cloudservicesdemo.in > Create record. The form is titled 'Record 3' and includes a 'Delete' button in the top right. The 'Record name' field contains 'learn' and the domain '.cloudservicesdemo.in' is shown to its right. Below this, there is a note: 'Keep blank to create a record for the root domain.' and an 'Alias' toggle which is currently off. The 'Value' field contains '3.3.3.3'. The 'Record type' dropdown is set to 'A - Routes traffic to an IPv4 address and some AWS resources'. The 'Routing policy' dropdown is set to 'IP-based'. The 'Health check ID - optional' field is empty with a search icon and the text 'Choose health check'. The 'Location' dropdown is set to '* (Default location)'. The 'Record ID' field contains 'R3'.

- Click **Create Records**.
- This step provides a default DNS response for all other source IP addresses.

Step 7 – Verify the IP-Based Routing Policy

- Open **Command Prompt**.
- **Run:**
 - nslookup learn.<your-domain-name>

```

Windows PowerShell 5.1
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PS C:\Users\HP> cd downloads
PS C:\Users\HP\downloads> nslookup
Default Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

> learn.cloudservicesdemo.in
Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address:  3.3.3.3

> learn.cloudservicesdemo.in
Server:  reliance.reliance
Address:  2405:201:681d:ea79::c0a8:1d01

Non-authoritative answer:
Name:    learn.cloudservicesdemo.in
Address:  3.3.3.3

```

- Verify that the returned IP address matches the configured source IP range.
- Alternatively, use **Test Record** in the Hosted Zone to verify the default routing response.

The screenshot shows the AWS Route 53 console for the hosted zone 'cloudservicesdemo.in'. The 'Test record' button is highlighted in red. Below it, the 'Records (5)' tab is active, displaying a table of DNS records.

Record name	Type	Routing	Differentiator	Alias	Value/Route traffic to	TTL (s)	Health	Evaluation
cloudservice...	NS	Simple	-	No	ns-1598.awsdns-07.co.uk, ns-16.awsdns-02.com, ns-925.awsdns-51.net, ns-1212.awsdns-23.org	1,72,800	-	-
cloudservice...	SOA	Simple	-	No	ns-1598.awsdns-07.co.uk a...	900	-	-
learn.clouds...	A	IP-based	Gujrat	No	1.1.1.1	300	-	-
learn.clouds...	A	IP-based	Maharashtra	No	2.2.2.2	300	-	-
learn.clouds...	A	IP-based	* (Default location)	No	3.3.3.3	300	-	-

- This step verifies that Route 53 returns different IP addresses based on the source IP address.

The screenshot shows the 'Test record' configuration page in the AWS Route 53 console. The 'Get response' button is highlighted in red. The page displays the response returned by Route 53 for the record 'learn' of type 'A'.

Response returned by Route 53
 Response from Route 53 based on the following options.

Hosted zone
 cloudservicesdemo.in

Record name
 learn

Record type
 A

DNS response code
 No Error

Protocol
 UDP

Response returned by Route 53
 3.3.3.3

Verification

- Successfully created a CIDR Collection.
- Added multiple CIDR locations.
- Configured IP-Based Routing Policy records.
- Associated DNS records with CIDR locations.
- Verified source IP-based DNS resolution using **nslookup** and **Test Record**.